

**WORK SHEET - 03**

**VECTOR & 3D**

**[SINGLE CORRECT CHOICE TYPE]**

**Q.1 to Q.10** has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 If  $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$  and  $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$  where  $a, b, c \in \{-2, -1, 0, 1, 2\}$ , then number of possible non-zero vectors  $\vec{V}_2$  perpendicular to  $\vec{V}_1$  is  
(A) 10 (B) 13 (C) 15 (D) 18
- Q.2 If a plane meets co-ordinate axes in A, B, C such that the centroid of the triangle is  $(1, k, k^2)$  then the least distance of plane from origin, is  
(A)  $\left| \frac{3k^2}{\sqrt{1+k+k^2}} \right|$  (B)  $\left| \frac{3k}{\sqrt{1+k+k^2}} \right|$  (C)  $\left| \frac{3k^2}{\sqrt{1+k^2+k^4}} \right|$  (D)  $\left| \frac{3k}{\sqrt{1+k^2+k^4}} \right|$
- Q.3 Let  $\hat{a}$  and  $\hat{c}$  be unit vectors at an angle  $\frac{\pi}{3}$  with each other. If  $(\hat{a} \times (\hat{b} \times \hat{c})) \cdot (\hat{a} \times \hat{c}) = 5$ , then the value of  $[\hat{a} \hat{b} \hat{c}]$  is equal to  
(A) 10 (B) -10 (C) 5 (D) -5
- Q.4 If  $\vec{a}$  and  $\vec{b}$  are non-zero vectors such that  $|\vec{a} + \vec{b}| = |\vec{a} - 2\vec{b}|$ , then  
(A)  $\vec{a} \cdot \vec{b} = 2|\vec{b}|^2$  (B)  $\vec{a} \cdot \vec{b} = |\vec{b}|^2$   
(C) least value of  $\vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|+2}$  is  $2\sqrt{2}$ . (D) least value of  $\vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|^2+2}$  is  $2\sqrt{2} - 1$ .
- Q.5  $P(1, 7, \sqrt{2})$  be a point and line L is  $2\sqrt{2}(x-1) = y-2, z=0$ . If PQ is distance of plane  $\sqrt{2}x + y - z = 1$  from P measured along a line inclined at an angle of  $45^\circ$  with L and is minimum then PQ is  
(A) 2 (B) 3 (C) 4 (D) 5
- Q.6 Let OABC be a regular tetrahedron of side length unity, where O is origin. If P is a point at a unit distance from origin such that OP is equally inclined to  $\vec{OA}$ ,  $\vec{OB}$  and  $\vec{OC}$  at an angle  $\alpha$ . Then  $\cos^2 \alpha$  equals  
(A) 0 (B)  $\frac{1}{3}$  (C)  $\frac{2}{3}$  (D)  $\frac{5}{6}$
- Q.7 Volume of parallelopiped whose adjacent sides are given by vectors  $\vec{\alpha}$ ,  $\vec{\beta}$  and  $\vec{\gamma}$ , where  $\vec{\alpha}$ ,  $\vec{\beta}$  and  $\vec{\gamma}$  are unit vectors which are bisecting the angle between three pair wise orthonormal vectors, is  
(A)  $\frac{1}{2\sqrt{2}}$  (B)  $\sqrt{2}$  (C)  $2\sqrt{2}$  (D)  $\frac{1}{\sqrt{2}}$

Q.8 If  $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ ,  $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$  and  $\vec{c} = \hat{i} + x\hat{j} + y\hat{k}$  are linearly dependent and  $|\vec{c}| = \sqrt{3}$ , then (x, y) is (are)

- (A)  $\left(\frac{-7}{5}, \frac{-1}{5}\right)$  (B)  $(-2, 0)$  (C)  $\left(\frac{1}{5}, \frac{-7}{5}\right)$  (D)  $(1, 1)$

Q.9 If  $\vec{\alpha}$  and  $\vec{\beta}$  are any two vectors of magnitudes 2 and 3 respectively such that  $2|\vec{\alpha} \times \vec{\beta}| + 3|\vec{\alpha} \cdot \vec{\beta}| = \gamma$ , then the maximum value of  $\gamma$ , is

- (A)  $\sqrt{13}$  (B)  $2\sqrt{13}$  (C)  $6\sqrt{13}$  (D)  $10\sqrt{13}$

Q.10 If  $\vec{p}$ ,  $\vec{q}$ ,  $\vec{r}$  are unit vectors such that  $\vec{p} \cdot \vec{q} = 0 = \vec{p} \cdot \vec{r}$  and the angle between  $\vec{q}$  and  $\vec{r}$  is  $\frac{\pi}{3}$ , then the value of  $|\vec{p} \times \vec{q} - \vec{p} \times \vec{r}|$  is equal to

- (A) 0 (B)  $\frac{1}{2}$  (C) 1 (D) 2

### [PARAGRAPH TYPE]

Q.11 to Q.13 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

#### Paragraph for question nos. 11 to 13

Let  $\vec{r}_1 = a^2\hat{i} - 2b\hat{j} - 6\hat{k}$  and  $\vec{r}_2 = 2a\hat{i} - a\hat{j} - a^2\hat{k}$  be non-zero vectors and  $\vec{r}_3 = \hat{i} - 2\hat{j} + a\hat{k}$  be a position vector of a point where  $a, b \in \mathbb{R}$ .  $P_1$  and  $P_2$  are two planes for the largest integral value of  $b$ , which containing vector  $\vec{r}_1$  and  $\vec{r}_2$  is the normal vector of both the planes and passing through point  $\vec{r}_3$ .

Q.11 If  $\theta$  is the acute angle between the planes then  $\cos \theta$  is

- (A) 0 (B)  $\frac{7}{3\sqrt{6}}$  (C)  $\frac{7}{18}$  (D) 1

Q.12 If  $d_1$  and  $d_2$  are perpendicular distances of the planes from the origin then  $(d_1 + d_2)$  equals

- (A) 0 (B)  $\frac{5}{\sqrt{6}}$  (C)  $\frac{7}{6\sqrt{6}}$  (D)  $\frac{3}{\sqrt{6}}$

Q.13 Equation of the plane containing the line  $P_1 = 0 = P_2$  and passing through the origin is

- (A)  $2x - y + 2z = 0$  (B)  $2x + y - z = 0$  (C)  $2x + y - 2z = 0$  (D)  $x + y - 2z = 0$

### [REASONING TYPE]

Q.14 to Q.16 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.14 Consider a plane  $P : x + 2y - 3z + 4 = 0$  and three lines

$$L_1 : \frac{x-1}{1} = \frac{y-3}{2} = \frac{2-z}{3}, L_2 : \frac{2x-1}{2} = \frac{y}{2} = \frac{z-1}{-3} \text{ and } L_3 : \frac{x-2}{2} = \frac{2y-10}{8} = \frac{z-5}{-6}$$

**Statement-1 :** Lines  $L_1$ ,  $L_2$  and  $L_3$  lie in the same plane.

**Statement-2 :** Lines  $L_1$ ,  $L_2$  and  $L_3$  are perpendicular to the plane  $P$ .

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

Q.15 **Statement-1 :** Let  $A(\vec{a})$ ,  $B(\vec{b})$  and  $C(\vec{c})$  be three points such that  $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ ,  $\vec{b} = 3\hat{i} - \hat{j} + 3\hat{k}$  and  $\vec{c} = -\hat{i} + 7\hat{j} - 5\hat{k}$ , then OABC is a tetrahedron.

**Statement-2 :** Let  $A(\vec{a})$ ,  $B(\vec{b})$  and  $C(\vec{c})$  be three points such that  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  are non-coplanar then OABC is a tetrahedron, where O is the origin.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (C) Statement-1 is true, statement-2 is false.  
 (D) Statement-1 is false, statement-2 is true.

Q.16 **Statement-1:** If  $\vec{b} \neq \vec{0}$ , then  $(\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$  represents a straight line passing through the point whose position vector is  $\vec{a}$  and the line is perpendicular to the plane  $\vec{r} \cdot \vec{b} = \vec{a} \cdot \vec{b}$ .

**Statement-2:** If  $\vec{b}$  is a unit vector and  $\vec{a} \cdot \vec{b} \neq 1$ , then  $\vec{r} \cdot \vec{b} = \vec{a} \cdot \vec{b}$  represents a plane passing through the point whose position vector is  $\vec{b}$ .

- (A) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1.  
 (B) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1.  
 (C) Statement-1 is true, Statement-2 is false.  
 (D) Statement-1 is false, Statement-2 is true.

### [MATRIX TYPE]

Q.17 has **four** statements (A, B, C, D) given in **Column-I** and **five** statements (P, Q, R, S, T) given in **Column-II**. Any given statement in **Column-I** can have correct matching with one or more statement(s) given in **Column-II**.

Q.17 Let P be a variable point in 3-D space and  $A(5, 0, -3)$ ,  $B(0, -5, 4)$  and  $C(0, 0, 0)$  be three points on the plane  $3x + 4y + 5z = 0$ . If the volume of the tetrahedron P-ABC is equal to 25 cubic units, then locus of P is  $ax + by + 5z \pm k = 0$ .

#### Column-I

- (A) The value of k is  
 (B) If A, B, C, D in order is a parallelogram, then area of the parallelogram is  
 (C) If G is the centroid of the  $\Delta ABC$ , then least value of PG is  
 (D) If H is the orthocentre of the  $\Delta ABC$ , then least value of PH is

#### Column-II

- (P)  $3\sqrt{2}$   
 (Q)  $15\sqrt{2}$   
 (R)  $25\sqrt{2}$   
 (S) 30  
 (T) 34

### [INTEGER TYPE]

Q.18 & Q.19 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.18 Let equation of plane be  $x + 2y + z - 3 = 0$ . An insect starts flying from point  $P(1, 3, 2)$  in straight line. It touches the plane at point  $R(a, b, c)$  and then goes to point  $Q(3, 5, 2)$  in straight line. If distance travelled  $PR + QR$  is minimum then find the value of  $(a + b + c)$ .

Q.19 Let  $\vec{a}$  and  $\vec{b}$  are two non-zero, non-collinear vectors ( $|\vec{a}| = 1$ ) such that vectors  $3(\vec{a} \times \vec{b})$  and  $2(\vec{b} - (\vec{a} \cdot \vec{b})\vec{a})$  represents two sides of a triangle. If area of the triangle is  $\frac{3}{4}(|\vec{b}|^4 + 4)$ , then find the value of  $[\vec{a}, \vec{b}]$ .

[Note :  $[k]$  denotes greatest integer less than or equal to k.]



## ANSWER KEY

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|      |   |      |                             |      |   |      |      |      |      |     |
|------|---|------|-----------------------------|------|---|------|------|------|------|-----|
| Q.1  | D | Q.2  | C                           | Q.3  | B | Q.4  | D    | Q.5  | B    |     |
| Q.6  | C | Q.7  | D                           | Q.8  | D | Q.9  | C    | Q.10 | C    |     |
| Q.11 | B | Q.12 | D                           | Q.13 | A | Q.14 | D    | Q.15 | D    |     |
| Q.16 | C | Q.17 | (A) S; (B) R; (C) P ; (D) P |      |   |      | Q.18 | [2]  | Q.19 | [1] |

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